

Pac-Man (pacman)

Alessandro is coding his new 3D version of Pac-Man. The game board is described by a three-dimensional grid, some cells are blocked and the others are free. Pac-Man and the ghosts can move to any free cell sharing a face with the current cell.

Alessandro instructed ghosts to move in a very simple way. If a ghost wants to move from cell A to cell B , it will repeat the following procedure until it reaches its destination or fails:

- If the ghost can decrease the distance along the x axis (i.e. $|A_x - B_x|$) it will do so by moving one cell along the x axis;
- otherwise if it can decrease the distance along the y axis (i.e. $|A_y - B_y|$) it will do so by moving one cell along the y axis;
- otherwise if it can decrease the distance along the z axis (i.e. $|A_z - B_z|$) it will do so by moving one cell along the z axis;
- otherwise it fails.

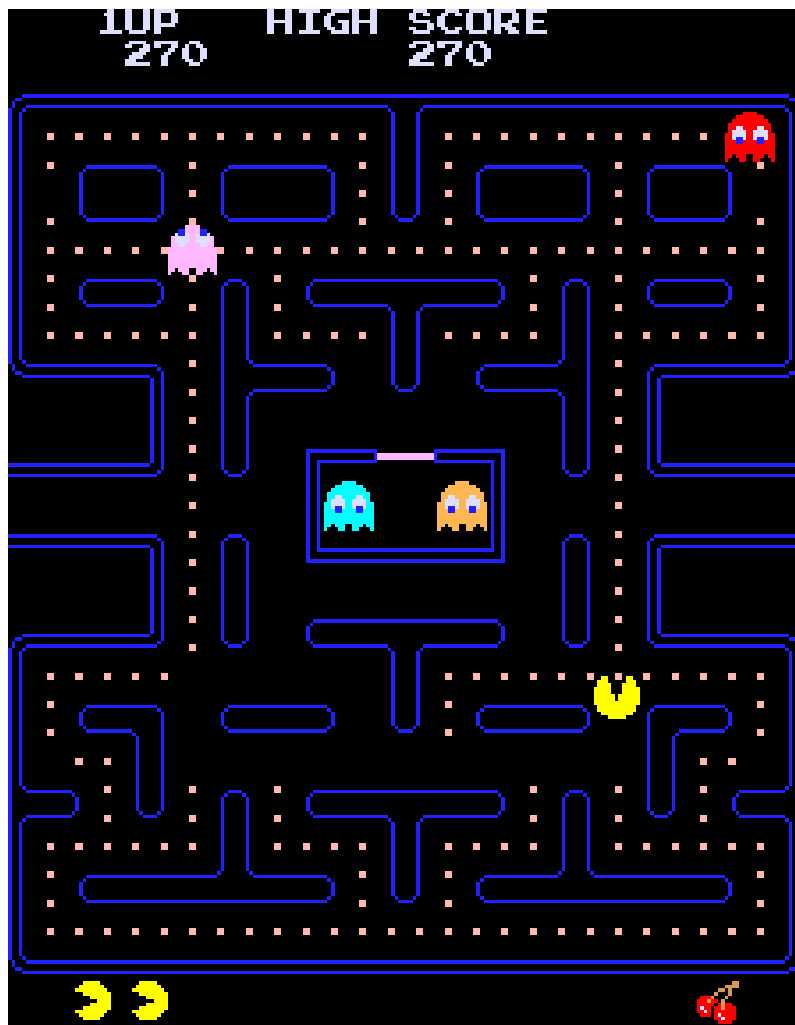


Figure 1: Alessandro playing a classic game of Pac-Man.

Alessandro has three arrays X_i , Y_i and Z_i indexed from 0 to $N - 1$ that describe the coordinates of the free cells. He is wondering whether the strategy above is smart enough to control the ghosts.

Help him by writing a program that determines whether for every pair of cells A and B a ghost will succeed in reaching cell B starting from cell A !

📎 Among the attachments of this task you may find a template file `pacman.*` with a sample incomplete implementation.

Input

The input file consists of:

- a line containing integer N .
- a line containing the N integers $X_0, \dots, X_{N-1}$.
- a line containing the N integers $Y_0, \dots, Y_{N-1}$.
- a line containing the N integers $Z_0, \dots, Z_{N-1}$.

Output

The output file must contain a single line consisting of **YES** if ghosts will always succeed in reaching their destinations or **NO** otherwise.

Constraints

- $1 \leq N \leq 100\,000$.
- $0 \leq X_i, Y_i, Z_i < 100\,000$ for each $i = 0 \dots N - 1$.

Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

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|---|---|
| – Subtask 1 (0 points) | Examples. |
|  | |
| – Subtask 2 (18 points) | $N \leq 100$ and $X_i, Y_i, Z_i < 100$. |
|  | |
| – Subtask 3 (19 points) | $N \leq 7500$ and $X_i, Y_i, Z_i < 100$. |
|  | |
| – Subtask 4 (24 points) | $N \leq 1000$ and $Z_i = 0$. |
|  | |
| – Subtask 5 (22 points) | $X_i, Y_i, Z_i < 100$. |
|  | |
| – Subtask 6 (17 points) | No additional limitations. |
|  | |

Examples

input	output
4 0 0 1 1 0 1 1 2 0 0 0 0	YES
8 0 1 2 2 2 1 0 0 0 0 0 1 1 1 1 0 0 0 0 0 1 1 1 1	NO
5 0 0 1 1 2 0 1 1 0 2 0 0 0 0 2	NO

Explanation

In the **first sample case**, ghosts can always reach their destination. For example, a ghost can move from cell $A = (0, 0, 0)$ to cell $B = (1, 2, 0)$ following path $(0, 0, 0) \rightarrow (0, 1, 0) \rightarrow (1, 1, 0) \rightarrow (1, 2, 0)$.

In the **second sample case**, ghost can't move from cell $A = (1, 0, 0)$ to cell $B = (1, 1, 1)$.